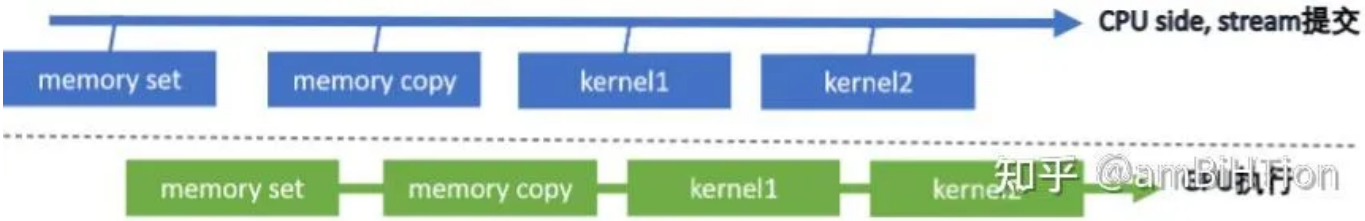
CUDA Stream

Stream 概念

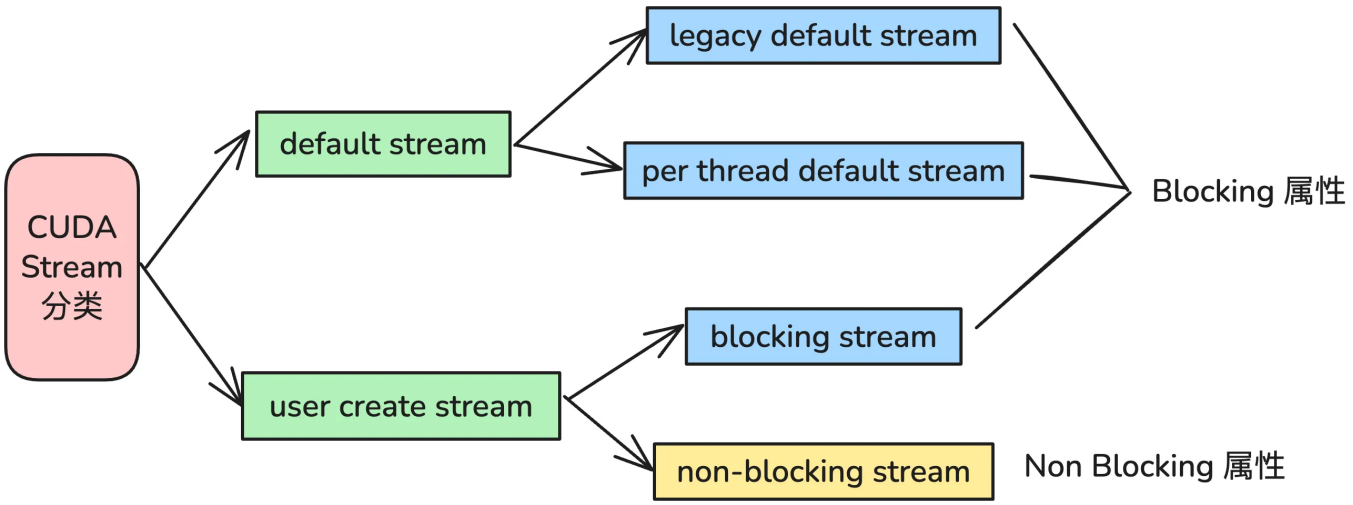
Stream 控制 GPU 侧任务异步执行的顺序，CUDA 运行时在 CPU 侧通过调用 Driver/Runtime API 的方式向 GPU 侧提交任务（主要是 kernel、memory 搬运及 graph），在提交任务时需要显示或隐式指定 stream，这保证了硬件执行顺序与 CPU 侧在同一 stream 上提交任务的顺序一致。



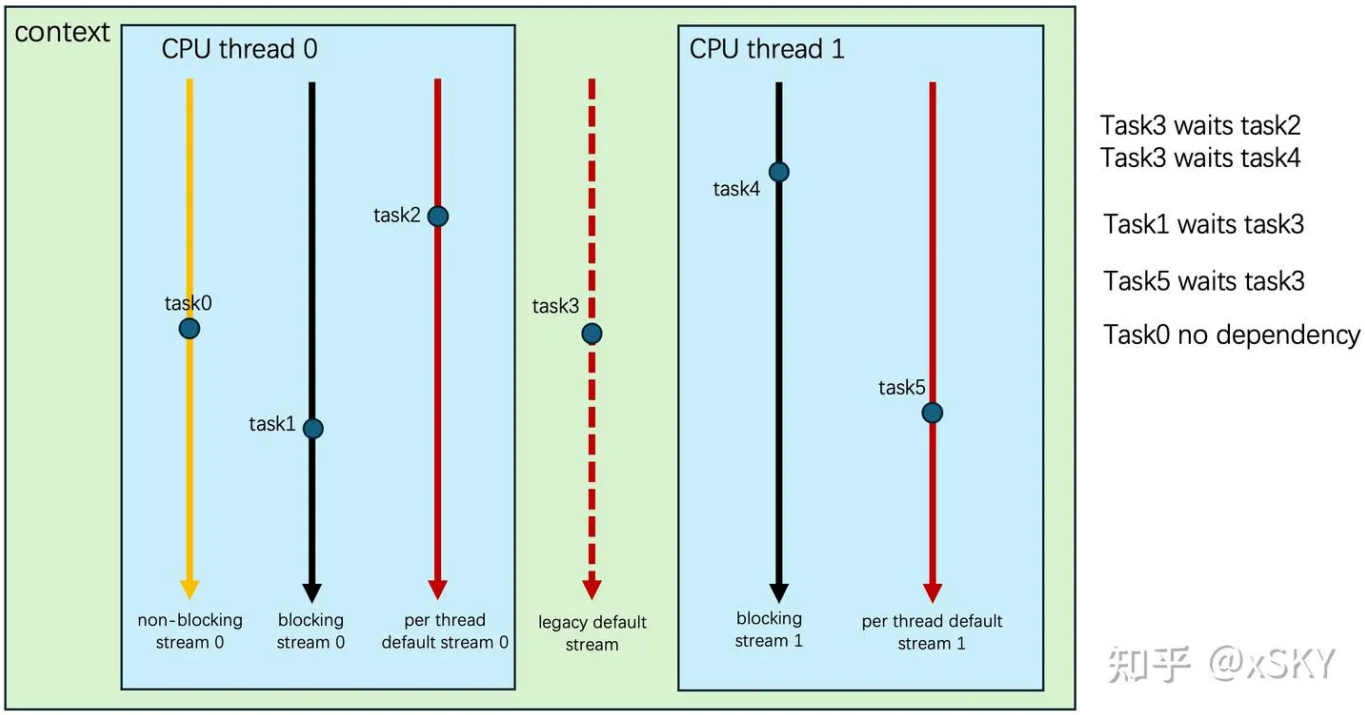
Stream 分类

总的来说，CUDA Stream 主要分为两类：`default stream` 和 `user create stream`。进一步细分，default stream 可分为 `legacy default stream` 和 `per thread default stream` 两类；user create stream 可分为 `blocking stream` 和 `non-blocking stream`。

所有不同种类的 stream 有两个互斥的属性 `Non Blocking` 和 `Blocking`，体现在 API 层面为：在 `cudaStreamCreateWithFlags` 传入 `cudaStreamNonBlocking` 或 `cudaStreamDefault`（这里不要与 default stream 搞混，应该理解为 `cudaStreamBlocking`）。**Non Blocking 和 Blocking 属性的语义为 GPU 执行时是否与 legacy default stream 添加隐式依赖关系**，blocking stream 提交的任务在 GPU 执行时要等待之前在 legacy default stream 上提交的任务做完再执行，legacy default stream 在执行时要等待所有具有 blocking 属性的其他 stream 先前提交的任务做完再执行。legacy default stream 具有很强的同步依赖属性，可以理解为一个“锚”。**Blocking stream 与 legacy default stream 在 GPU 执行时互相阻塞，依赖顺序与 CPU 侧提交的时间顺序一致。**



再来区分下 legacy default stream 和 per thread default stream，CUDA stream 的管理由 CUDA context 负责，每个 context 有一条 legacy default stream，每个 context 每个 CPU 线程有一条 per thread default stream。因此**在不同线程里使用 legacy default stream 是同一条，而在不同线程里使用的 per thread default stream 是不同的**。除此之外，**per thread default stream 具备 blocking 属性**，需要与 legacy default stream 添加上一段描述的隐式依赖。



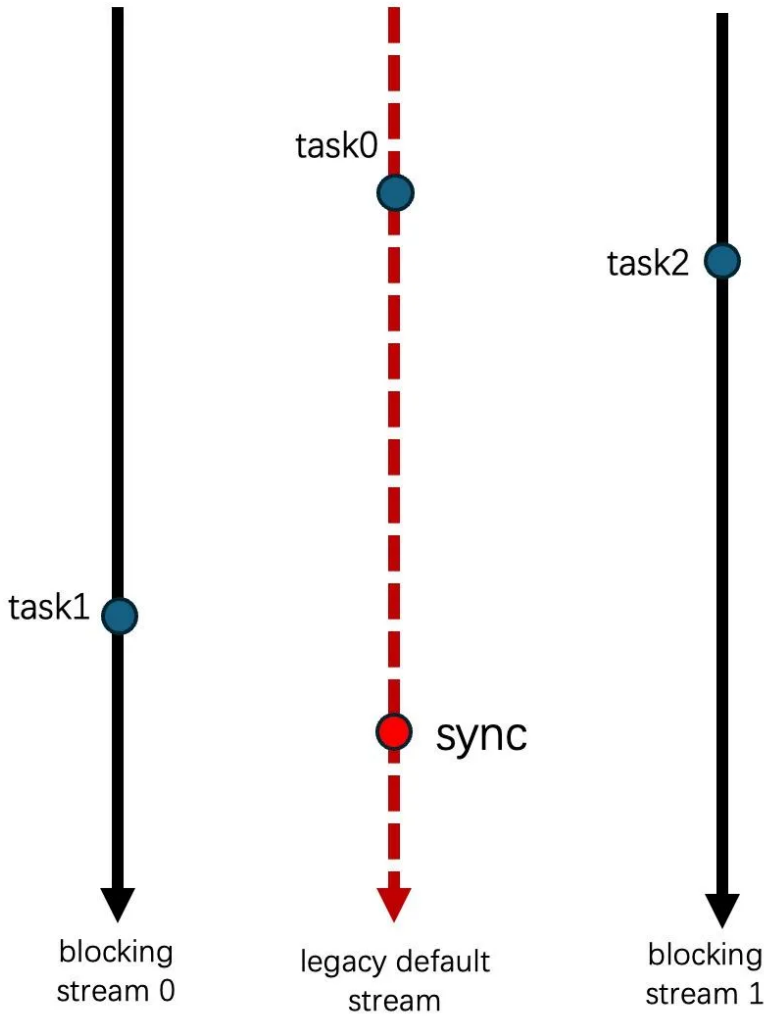
Default stream 的使用

使用驱动提供的 default stream 需要使用三个特殊的 stream handle 作为参数，分别为 `0，1，2` 。**1 表示 legacy default stream，2 表示 per thread default stream**。0 根据编译条件驱动内部转化为 1 或 2，带着 `--default-stream CUDA_API_PER_THREAD_DEFAULT_STREAM`（或添加宏）编译，传入的 handle 0 转为 2，不带这个参数编译传入的 handle 0 转为 1。**不带参数的同步 API 默认选择参数 0 作为 stream handle，遵循上述规则做转换。**

Stream 同步

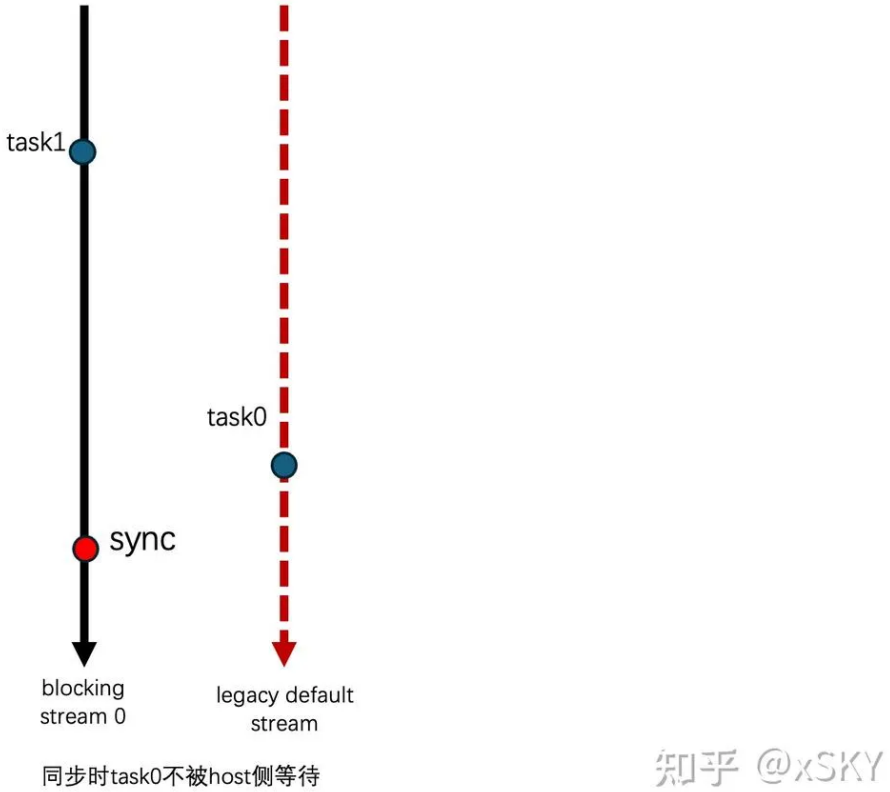
对于 stream 同步，API 有 `cudaStreamSynchronize` 和 `cuStreamSynchronize`，如下图 3 和图 4 所示。同步某个 stream 时，CPU 除了会等待该 stream 上提交的任务完成，还有可能会等待与当前 stream 有隐式依赖的其他 stream 上提交的任务完成，即：**同步 legacy default stream 除同步其自身，所有其他具有 blocking 属性的 stream（per thread default stream 和 blocking stream）都被同步，而同步其他 blocking stream，却并不会同步 legacy default stream。**只有将同步 stream 改为在 legacy default stream 上记录一个 event，并且同步这个 event 时才会显示的同步依赖。

> 注意：被同步的 blocking stream 或 legacy default stream 原本就会互相强依赖，即被同步 stream 最近一次提交之前在隐式依赖上提交的任务本就会被同步。所以上述表述的区别在于，同步 legacy default stream 会同步被同步 stream 最近一次提交后及此次调用同步 API 前，有隐式依赖的其他 stream 上提交的新任务；反之，同步 blocking stream 并不会同步 legacy default stream 上的这部分新任务。



同步时task1和task2也被host侧等待

知乎 @xSKY



不同种类的 stream 在 graph capture 时也有限制，legacy default stream 不能被 capture，其他所有stream 均可以被 capture，这是因为 legacy default stream 的依赖关系太强，如果可以被 capture 会把所有其他 blocking stream 拉入 capture 状态，从实现的角度不能被 capture 是合理的。因此在 blocking stream capture 阶段不允许向 legacy default stream 提交新任务以保证 capture 的逻辑正确性。

总结

- 如果在编译时未指定相关 flag，或指定 `--default-stream legacy`，则默认流是一个特殊的流，称作 NULL stream。同一个设备的主机线程会共享这个 NULL stream。NULL stream 是一个同步流，所有命令会产生隐式的同步。对于未指定 `--default-stream` 进行编译的代码，将 `--default-stream legacy` 视为默认值。
- 如果在编译时使用了 `--default-stream per-thread`，或是在 include 任何 CUDA 头文件（包括 cuda.h 和 cuda_runtime.h）前定义 `CUDA_API_PER_THREAD_DEFAULT_STREAM 1` 这个宏，则主机端的每一个线程都有自己专属的默认流，即 per-thread default stream。
- 注意：当代码由 nvcc 编译时，`#define CUDA_API_PER_THREAD_DEFAULT_STREAM`

`1` 不能用于启用每个主机线程的默认流，因为 nvcc 在编译时会在每个编译单元（.cu 文件）之前隐式包含 cuda_runtime.h。在这种情况下，需要使用 `--default-stream per-thread` 编译标志，或者需要使用 `-DCUDA_API_PER_THREAD_DEFAULT_STREAM=1` 编译宏。

- `--default-stream per-thread` 编译选项必须在一开始编译的时候就加上，如果调用的 lib 里面有 cuda 代码，那么编译库的时候就要加上该选项，否则就无法保证在最终的程序中生效。由于 legacy default stream 会隐式同步每个线程的内核函数调用，而我们又无法保证我们使用的库是否在编译的时候开启了 `--default-stream per-thread` 编译选项，所以最佳实践是显式创建 non-blocking stream：`cudaStreamCreateWithFlags(&stream, cudaStreamNonBlocking)`。

示例

1. default stream

1.1. legacy default stream

`cmake -B build && cmake --build build`

`./build/csrc/cuda_stream/default_stream`

result Bash

```bash
$./build/csrc/cuda_stream/default_stream
Legacy default stream
Starting two threads...
Using default stream handle: 0
Behavior depends on compilation flag.

Thread 0 started.
Clock Rate: 1980000000 Hz
GPU Cycle=59378 | Elapsed=0.00s | Thread 0 Step 1
GPU Cycle=1980143829 | Elapsed=1.00s | Thread 0 Step 2
GPU Cycle=3960228922 | Elapsed=2.00s | Thread 0 Step 3
GPU Cycle=5940313814 | Elapsed=3.00s | Thread 0 Step 4
GPU Cycle=7920399572 | Elapsed=4.00s | Thread 0 Step 5
GPU Cycle=9900485919 | Elapsed=5.00s | Thread 0 Step 6
GPU Cycle=11880570926 | Elapsed=6.00s | Thread 0 Step 7
GPU Cycle=13860656735 | Elapsed=7.00s | Thread 0 Step 8
GPU Cycle=15840741034 | Elapsed=8.00s | Thread 0 Step 9
GPU Cycle=17820826629 | Elapsed=9.00s | Thread 0 Step 10
Thread 1 started.
Thread 0 finished.

Clock Rate: 1980000000 Hz
GPU Cycle=60670 | Elapsed=0.00s | Thread 1 Step 1
GPU Cycle=1980146158 | Elapsed=1.00s | Thread 1 Step 2
GPU Cycle=3960231578 | Elapsed=2.00s | Thread 1 Step 3
GPU Cycle=5940317529 | Elapsed=3.00s | Thread 1 Step 4
GPU Cycle=7920403086 | Elapsed=4.00s | Thread 1 Step 5
GPU Cycle=9900488471 | Elapsed=5.00s | Thread 1 Step 6
GPU Cycle=11880574713 | Elapsed=6.00s | Thread 1 Step 7
GPU Cycle=13860660015 | Elapsed=7.00s | Thread 1 Step 8
GPU Cycle=15840745597 | Elapsed=8.00s | Thread 1 Step 9
GPU Cycle=17820831551 | Elapsed=9.00s | Thread 1 Step 10
Thread 1 finished.
```

1.2. per thread default stream

`cmake -B build -DUSE_PER_THREAD_STREAM=ON && cmake --build build`

`./build/csrc/cuda_stream/default_stream`

result C++

```
$./build/csrc/cuda_stream/default_stream
Per-thread default stream
Starting two threads...
Using default stream handle: 0
Behavior depends on compilation flag.

Thread 0 started.
Thread 1 started.
Clock Rate: 1980000000 Hz
Clock Rate: 1980000000 Hz
GPU Cycle=59157 | Elapsed=0.00s | Thread 0 Step 1
GPU Cycle=62835 | Elapsed=0.00s | Thread 1 Step 1
GPU Cycle=1980152169 | Elapsed=1.00s | Thread 1 Step 2
GPU Cycle=1980143582 | Elapsed=1.00s | Thread 0 Step 2
GPU Cycle=3960241362 | Elapsed=2.00s | Thread 1 Step 3
GPU Cycle=3960227125 | Elapsed=2.00s | Thread 0 Step 3
GPU Cycle=5940330549 | Elapsed=3.00s | Thread 1 Step 4
GPU Cycle=5940310943 | Elapsed=3.00s | Thread 0 Step 4
GPU Cycle=7920419891 | Elapsed=4.00s | Thread 1 Step 5
GPU Cycle=7920395507 | Elapsed=4.00s | Thread 0 Step 5
GPU Cycle=9900509389 | Elapsed=5.00s | Thread 1 Step 6
GPU Cycle=9900479722 | Elapsed=5.00s | Thread 0 Step 6
GPU Cycle=11880598255 | Elapsed=6.00s | Thread 1 Step 7
GPU Cycle=11880563644 | Elapsed=6.00s | Thread 0 Step 7
GPU Cycle=13860687206 | Elapsed=7.00s | Thread 1 Step 8
GPU Cycle=13860647695 | Elapsed=7.00s | Thread 0 Step 8
GPU Cycle=15840776779 | Elapsed=8.00s | Thread 1 Step 9
GPU Cycle=15840732362 | Elapsed=8.00s | Thread 0 Step 9
GPU Cycle=17820866399 | Elapsed=9.00s | Thread 1 Step 10
GPU Cycle=17820817088 | Elapsed=9.00s | Thread 0 Step 10
Thread 1 finished.

Thread 0 finished.
```

2. blocking stream

2.1. legacy default stream

`cmake -B build && cmake --build build`

`./build/csrc/cuda_stream/blocking_stream`

result C++

```
$./build/csrc/cuda_stream/blocking_stream
Legacy default stream
Starting two threads...
Using default stream handle: 0
Behavior depends on compilation flag.

Blocking stream 与 default stream
Kernel 0 started.
Clock Rate: 1980000000 Hz
GPU Cycle=60058 | Elapsed=0.00s | Kernel 0 Step 1
GPU Cycle=1980145658 | Elapsed=1.00s | Kernel 0 Step 2
GPU Cycle=3960230806 | Elapsed=2.00s | Kernel 0 Step 3
GPU Cycle=5940316131 | Elapsed=3.00s | Kernel 0 Step 4
GPU Cycle=7920401678 | Elapsed=4.00s | Kernel 0 Step 5
GPU Cycle=9900487396 | Elapsed=5.00s | Kernel 0 Step 6
GPU Cycle=11880572736 | Elapsed=6.00s | Kernel 0 Step 7
GPU Cycle=13860657961 | Elapsed=7.00s | Kernel 0 Step 8
GPU Cycle=15840743413 | Elapsed=8.00s | Kernel 0 Step 9
GPU Cycle=17820828526 | Elapsed=9.00s | Kernel 0 Step 10
Kernel 1 started.
Clock Rate: 1980000000 Hz
GPU Cycle=59405 | Elapsed=0.00s | Kernel 1 Step 1
GPU Cycle=1980144910 | Elapsed=1.00s | Kernel 1 Step 2
GPU Cycle=3960230217 | Elapsed=2.00s | Kernel 1 Step 3
GPU Cycle=5940315293 | Elapsed=3.00s | Kernel 1 Step 4
GPU Cycle=7920400807 | Elapsed=4.00s | Kernel 1 Step 5
GPU Cycle=9900486148 | Elapsed=5.00s | Kernel 1 Step 6
GPU Cycle=11880571814 | Elapsed=6.00s | Kernel 1 Step 7
GPU Cycle=13860657259 | Elapsed=7.00s | Kernel 1 Step 8
GPU Cycle=15840742551 | Elapsed=8.00s | Kernel 1 Step 9
GPU Cycle=17820828363 | Elapsed=9.00s | Kernel 1 Step 10
Kernel 2 started.
Clock Rate: 1980000000 Hz
GPU Cycle=59929 | Elapsed=0.00s | Kernel 2 Step 1
GPU Cycle=1980145369 | Elapsed=1.00s | Kernel 2 Step 2
GPU Cycle=3960230666 | Elapsed=2.00s | Kernel 2 Step 3
GPU Cycle=5940315910 | Elapsed=3.00s | Kernel 2 Step 4
GPU Cycle=7920401317 | Elapsed=4.00s | Kernel 2 Step 5
GPU Cycle=9900487089 | Elapsed=5.00s | Kernel 2 Step 6
GPU Cycle=11880572327 | Elapsed=6.00s | Kernel 2 Step 7
GPU Cycle=13860657847 | Elapsed=7.00s | Kernel 2 Step 8
GPU Cycle=15840743201 | Elapsed=8.00s | Kernel 2 Step 9
GPU Cycle=17820829604 | Elapsed=9.00s | Kernel 2 Step 10
Blocking stream 与 default stream 完成
Non-blocking stream 与 default stream
```

```
Kernel 0 started.
Kernel 1 started.
Clock Rate: 1980000000 Hz
Clock Rate: 1980000000 Hz
GPU Cycle=60884 | Elapsed=0.00s | Kernel 0 Step 1
GPU Cycle=61910 | Elapsed=0.00s | Kernel 1 Step 1
GPU Cycle=1980151364 | Elapsed=1.00s | Kernel 1 Step 2
GPU Cycle=1980145884 | Elapsed=1.00s | Kernel 0 Step 2
GPU Cycle=3960240765 | Elapsed=2.00s | Kernel 1 Step 3
GPU Cycle=3960231149 | Elapsed=2.00s | Kernel 0 Step 3
GPU Cycle=5940330017 | Elapsed=3.00s | Kernel 1 Step 4
GPU Cycle=5940316765 | Elapsed=3.00s | Kernel 0 Step 4
GPU Cycle=7920420440 | Elapsed=4.00s | Kernel 1 Step 5
GPU Cycle=7920403284 | Elapsed=4.00s | Kernel 0 Step 5
GPU Cycle=9900510506 | Elapsed=5.00s | Kernel 1 Step 6
GPU Cycle=9900488892 | Elapsed=5.00s | Kernel 0 Step 6
GPU Cycle=11880599985 | Elapsed=6.00s | Kernel 1 Step 7
GPU Cycle=11880574422 | Elapsed=6.00s | Kernel 0 Step 7
GPU Cycle=13860689816 | Elapsed=7.00s | Kernel 1 Step 8
GPU Cycle=13860660204 | Elapsed=7.00s | Kernel 0 Step 8
GPU Cycle=15840779472 | Elapsed=8.00s | Kernel 1 Step 9
GPU Cycle=15840746406 | Elapsed=8.00s | Kernel 0 Step 9
GPU Cycle=17820868956 | Elapsed=9.00s | Kernel 1 Step 10
GPU Cycle=17820832401 | Elapsed=9.00s | Kernel 0 Step 10
Kernel 2 started.
Clock Rate: 1980000000 Hz
GPU Cycle=59223 | Elapsed=0.00s | Kernel 2 Step 1
GPU Cycle=1980143340 | Elapsed=1.00s | Kernel 2 Step 2
GPU Cycle=3960229276 | Elapsed=2.00s | Kernel 2 Step 3
GPU Cycle=5940314677 | Elapsed=3.00s | Kernel 2 Step 4
GPU Cycle=7920400437 | Elapsed=4.00s | Kernel 2 Step 5
GPU Cycle=9900486101 | Elapsed=5.00s | Kernel 2 Step 6
GPU Cycle=11880571451 | Elapsed=6.00s | Kernel 2 Step 7
GPU Cycle=13860657028 | Elapsed=7.00s | Kernel 2 Step 8
GPU Cycle=15840742771 | Elapsed=8.00s | Kernel 2 Step 9
GPU Cycle=17820828651 | Elapsed=9.00s | Kernel 2 Step 10
Non-blocking stream 与 default stream 完成
Default stream 与 blocking stream
Kernel 0 started.
Clock Rate: 1980000000 Hz
GPU Cycle=59365 | Elapsed=0.00s | Kernel 0 Step 1
GPU Cycle=1980143990 | Elapsed=1.00s | Kernel 0 Step 2
GPU Cycle=3960229679 | Elapsed=2.00s | Kernel 0 Step 3
GPU Cycle=5940315340 | Elapsed=3.00s | Kernel 0 Step 4
GPU Cycle=7920400633 | Elapsed=4.00s | Kernel 0 Step 5
GPU Cycle=9900486187 | Elapsed=5.00s | Kernel 0 Step 6
GPU Cycle=11880571456 | Elapsed=6.00s | Kernel 0 Step 7
GPU Cycle=13860657020 | Elapsed=7.00s | Kernel 0 Step 8
```

```
GPU Cycle=15840742606 | Elapsed=8.00s | Kernel 0 Step 9
GPU Cycle=17820828126 | Elapsed=9.00s | Kernel 0 Step 10
Kernel 1 started.
Clock Rate: 1980000000 Hz
GPU Cycle=59086 | Elapsed=0.00s | Kernel 1 Step 1
GPU Cycle=1980143336 | Elapsed=1.00s | Kernel 1 Step 2
GPU Cycle=3960229288 | Elapsed=2.00s | Kernel 1 Step 3
GPU Cycle=5940314652 | Elapsed=3.00s | Kernel 1 Step 4
GPU Cycle=7920400202 | Elapsed=4.00s | Kernel 1 Step 5
GPU Cycle=9900485785 | Elapsed=5.00s | Kernel 1 Step 6
GPU Cycle=11880571571 | Elapsed=6.00s | Kernel 1 Step 7
GPU Cycle=13860657127 | Elapsed=7.00s | Kernel 1 Step 8
GPU Cycle=15840742601 | Elapsed=8.00s | Kernel 1 Step 9
GPU Cycle=17820828319 | Elapsed=9.00s | Kernel 1 Step 10
Kernel 2 started.
Clock Rate: 1980000000 Hz
GPU Cycle=59211 | Elapsed=0.00s | Kernel 2 Step 1
GPU Cycle=1980143455 | Elapsed=1.00s | Kernel 2 Step 2
GPU Cycle=3960228628 | Elapsed=2.00s | Kernel 2 Step 3
GPU Cycle=5940314788 | Elapsed=3.00s | Kernel 2 Step 4
GPU Cycle=7920400327 | Elapsed=4.00s | Kernel 2 Step 5
GPU Cycle=9900486400 | Elapsed=5.00s | Kernel 2 Step 6
GPU Cycle=11880572318 | Elapsed=6.00s | Kernel 2 Step 7
GPU Cycle=13860658402 | Elapsed=7.00s | Kernel 2 Step 8
GPU Cycle=15840744469 | Elapsed=8.00s | Kernel 2 Step 9
GPU Cycle=17820830680 | Elapsed=9.00s | Kernel 2 Step 10
Default stream 与 blocking stream 完成
Default stream 与 non-blocking stream
Kernel 0 started.
Kernel 1 started.
Clock Rate: 1980000000 Hz
Clock Rate: 1980000000 Hz
GPU Cycle=59382 | Elapsed=0.00s | Kernel 0 Step 1
GPU Cycle=62099 | Elapsed=0.00s | Kernel 1 Step 1
GPU Cycle=1980150922 | Elapsed=1.00s | Kernel 1 Step 2
GPU Cycle=1980142970 | Elapsed=1.00s | Kernel 0 Step 2
GPU Cycle=3960240165 | Elapsed=2.00s | Kernel 1 Step 3
GPU Cycle=3960228236 | Elapsed=2.00s | Kernel 0 Step 3
GPU Cycle=5940329651 | Elapsed=3.00s | Kernel 1 Step 4
GPU Cycle=5940313707 | Elapsed=3.00s | Kernel 0 Step 4
GPU Cycle=7920419025 | Elapsed=4.00s | Kernel 1 Step 5
GPU Cycle=7920399194 | Elapsed=4.00s | Kernel 0 Step 5
GPU Cycle=9900508280 | Elapsed=5.00s | Kernel 1 Step 6
GPU Cycle=9900484665 | Elapsed=5.00s | Kernel 0 Step 6
GPU Cycle=11880597982 | Elapsed=6.00s | Kernel 1 Step 7
GPU Cycle=11880570004 | Elapsed=6.00s | Kernel 0 Step 7
GPU Cycle=13860686981 | Elapsed=7.00s | Kernel 1 Step 8
GPU Cycle=13860655328 | Elapsed=7.00s | Kernel 0 Step 8
```

```
GPU Cycle=15840776352 | Elapsed=8.00s | Kernel 1 Step 9
GPU Cycle=15840740839 | Elapsed=8.00s | Kernel 0 Step 9
GPU Cycle=17820865800 | Elapsed=9.00s | Kernel 1 Step 10
GPU Cycle=17820826665 | Elapsed=9.00s | Kernel 0 Step 10
Kernel 2 started.
Clock Rate: 1980000000 Hz
GPU Cycle=59362 | Elapsed=0.00s | Kernel 2 Step 1
GPU Cycle=1980143831 | Elapsed=1.00s | Kernel 2 Step 2
GPU Cycle=3960229760 | Elapsed=2.00s | Kernel 2 Step 3
GPU Cycle=5940315134 | Elapsed=3.00s | Kernel 2 Step 4
GPU Cycle=7920401181 | Elapsed=4.00s | Kernel 2 Step 5
GPU Cycle=9900487181 | Elapsed=5.00s | Kernel 2 Step 6
GPU Cycle=11880572861 | Elapsed=6.00s | Kernel 2 Step 7
GPU Cycle=13860658688 | Elapsed=7.00s | Kernel 2 Step 8
GPU Cycle=15840744655 | Elapsed=8.00s | Kernel 2 Step 9
GPU Cycle=17820830448 | Elapsed=9.00s | Kernel 2 Step 10
Default stream 与 non-blocking stream 完成
Non-blocking stream 与 blocking stream
Kernel 0 started.
Kernel 1 started.
Clock Rate: 1980000000 Hz
Clock Rate: 1980000000 Hz
GPU Cycle=60346 | Elapsed=0.00s | Kernel 0 Step 1
GPU Cycle=62030 | Elapsed=0.00s | Kernel 1 Step 1
GPU Cycle=1980150775 | Elapsed=1.00s | Kernel 1 Step 2
GPU Cycle=1980144669 | Elapsed=1.00s | Kernel 0 Step 2
GPU Cycle=3960239614 | Elapsed=2.00s | Kernel 1 Step 3
GPU Cycle=3960228965 | Elapsed=2.00s | Kernel 0 Step 3
GPU Cycle=5940330567 | Elapsed=3.00s | Kernel 1 Step 4
GPU Cycle=5940314589 | Elapsed=3.00s | Kernel 0 Step 4
GPU Cycle=7920420243 | Elapsed=4.00s | Kernel 1 Step 5
GPU Cycle=7920400804 | Elapsed=4.00s | Kernel 0 Step 5
GPU Cycle=9900509189 | Elapsed=5.00s | Kernel 1 Step 6
GPU Cycle=9900486267 | Elapsed=5.00s | Kernel 0 Step 6
GPU Cycle=11880598677 | Elapsed=6.00s | Kernel 1 Step 7
GPU Cycle=11880571730 | Elapsed=6.00s | Kernel 0 Step 7
GPU Cycle=13860687642 | Elapsed=7.00s | Kernel 1 Step 8
GPU Cycle=13860656957 | Elapsed=7.00s | Kernel 0 Step 8
GPU Cycle=15840776936 | Elapsed=8.00s | Kernel 1 Step 9
GPU Cycle=15840742537 | Elapsed=8.00s | Kernel 0 Step 9
GPU Cycle=17820866449 | Elapsed=9.00s | Kernel 1 Step 10
GPU Cycle=17820828684 | Elapsed=9.00s | Kernel 0 Step 10
Kernel 2 started.
Clock Rate: 1980000000 Hz
GPU Cycle=59426 | Elapsed=0.00s | Kernel 2 Step 1
GPU Cycle=1980143910 | Elapsed=1.00s | Kernel 2 Step 2
GPU Cycle=3960230342 | Elapsed=2.00s | Kernel 2 Step 3
GPU Cycle=5940316504 | Elapsed=3.00s | Kernel 2 Step 4
```

```
GPU Cycle=7920403120 | Elapsed=4.00s | Kernel 2 Step 5
GPU Cycle=9900489329 | Elapsed=5.00s | Kernel 2 Step 6
GPU Cycle=11880574754 | Elapsed=6.00s | Kernel 2 Step 7
GPU Cycle=13860661176 | Elapsed=7.00s | Kernel 2 Step 8
GPU Cycle=15840747079 | Elapsed=8.00s | Kernel 2 Step 9
GPU Cycle=17820833024 | Elapsed=9.00s | Kernel 2 Step 10
Non-blocking stream 与 blocking stream 完成
Thread 0 finished.
```

2.2. per thread default stream

`cmake -B build -DUSE_PER_THREAD_STREAM=ON && cmake --build build`

`./build/csrc/cuda_stream/blocking_stream`

result C++

```
$./build/csrc/cuda_stream/blocking_stream
Per-thread default stream
Starting two threads...
Using default stream handle: 0
Behavior depends on compilation flag.

Blocking stream 与 default stream
Kernel 0 started.
Clock Rate: 1980000000 Hz
Kernel 1 started.
GPU Cycle=59412 | Elapsed=0.00s | Kernel 0 Step 1
Clock Rate: 1980000000 Hz
GPU Cycle=61207 | Elapsed=0.00s | Kernel 1 Step 1
GPU Cycle=1980149627 | Elapsed=1.00s | Kernel 1 Step 2
GPU Cycle=1980143675 | Elapsed=1.00s | Kernel 0 Step 2
GPU Cycle=3960237926 | Elapsed=2.00s | Kernel 1 Step 3
GPU Cycle=3960228523 | Elapsed=2.00s | Kernel 0 Step 3
GPU Cycle=5940326204 | Elapsed=3.00s | Kernel 1 Step 4
GPU Cycle=5940312795 | Elapsed=3.00s | Kernel 0 Step 4
GPU Cycle=7920414659 | Elapsed=4.00s | Kernel 1 Step 5
GPU Cycle=7920397360 | Elapsed=4.00s | Kernel 0 Step 5
GPU Cycle=9900503327 | Elapsed=5.00s | Kernel 1 Step 6
GPU Cycle=9900482275 | Elapsed=5.00s | Kernel 0 Step 6
GPU Cycle=11880591547 | Elapsed=6.00s | Kernel 1 Step 7
GPU Cycle=11880566632 | Elapsed=6.00s | Kernel 0 Step 7
GPU Cycle=13860679915 | Elapsed=7.00s | Kernel 1 Step 8
GPU Cycle=13860651090 | Elapsed=7.00s | Kernel 0 Step 8
GPU Cycle=15840768407 | Elapsed=8.00s | Kernel 1 Step 9
GPU Cycle=15840735677 | Elapsed=8.00s | Kernel 0 Step 9
GPU Cycle=17820856574 | Elapsed=9.00s | Kernel 1 Step 10
GPU Cycle=17820820002 | Elapsed=9.00s | Kernel 0 Step 10
Kernel 2 started.
Clock Rate: 1980000000 Hz
GPU Cycle=58325 | Elapsed=0.00s | Kernel 2 Step 1
GPU Cycle=1980142135 | Elapsed=1.00s | Kernel 2 Step 2
GPU Cycle=3960226968 | Elapsed=2.00s | Kernel 2 Step 3
GPU Cycle=5940311206 | Elapsed=3.00s | Kernel 2 Step 4
GPU Cycle=7920395984 | Elapsed=4.00s | Kernel 2 Step 5
GPU Cycle=9900480604 | Elapsed=5.00s | Kernel 2 Step 6
GPU Cycle=11880564920 | Elapsed=6.00s | Kernel 2 Step 7
GPU Cycle=13860649683 | Elapsed=7.00s | Kernel 2 Step 8
GPU Cycle=15840734107 | Elapsed=8.00s | Kernel 2 Step 9
GPU Cycle=17820818708 | Elapsed=9.00s | Kernel 2 Step 10
Blocking stream 与 default stream 完成
Non-blocking stream 与 default stream
```

```
Kernel 0 started.
Kernel 1 started.
Clock Rate: 1980000000 Hz
Clock Rate: 1980000000 Hz
GPU Cycle=58309 | Elapsed=0.00s | Kernel 0 Step 1
GPU Cycle=61544 | Elapsed=0.00s | Kernel 1 Step 1
GPU Cycle=1980149548 | Elapsed=1.00s | Kernel 1 Step 2
GPU Cycle=1980142775 | Elapsed=1.00s | Kernel 0 Step 2
GPU Cycle=3960238126 | Elapsed=2.00s | Kernel 1 Step 3
GPU Cycle=3960227205 | Elapsed=2.00s | Kernel 0 Step 3
GPU Cycle=5940325975 | Elapsed=3.00s | Kernel 1 Step 4
GPU Cycle=5940311363 | Elapsed=3.00s | Kernel 0 Step 4
GPU Cycle=7920414765 | Elapsed=4.00s | Kernel 1 Step 5
GPU Cycle=7920395713 | Elapsed=4.00s | Kernel 0 Step 5
GPU Cycle=9900503311 | Elapsed=5.00s | Kernel 1 Step 6
GPU Cycle=9900480414 | Elapsed=5.00s | Kernel 0 Step 6
GPU Cycle=11880591680 | Elapsed=6.00s | Kernel 1 Step 7
GPU Cycle=11880565032 | Elapsed=6.00s | Kernel 0 Step 7
GPU Cycle=13860679826 | Elapsed=7.00s | Kernel 1 Step 8
GPU Cycle=13860649258 | Elapsed=7.00s | Kernel 0 Step 8
GPU Cycle=15840768165 | Elapsed=8.00s | Kernel 1 Step 9
GPU Cycle=15840733630 | Elapsed=8.00s | Kernel 0 Step 9
GPU Cycle=17820856511 | Elapsed=9.00s | Kernel 1 Step 10
GPU Cycle=17820818252 | Elapsed=9.00s | Kernel 0 Step 10
Kernel 2 started.
Clock Rate: 1980000000 Hz
GPU Cycle=58217 | Elapsed=0.00s | Kernel 2 Step 1
GPU Cycle=1980142232 | Elapsed=1.00s | Kernel 2 Step 2
GPU Cycle=3960226836 | Elapsed=2.00s | Kernel 2 Step 3
GPU Cycle=5940311272 | Elapsed=3.00s | Kernel 2 Step 4
GPU Cycle=7920395747 | Elapsed=4.00s | Kernel 2 Step 5
GPU Cycle=9900480054 | Elapsed=5.00s | Kernel 2 Step 6
GPU Cycle=11880565084 | Elapsed=6.00s | Kernel 2 Step 7
GPU Cycle=13860650080 | Elapsed=7.00s | Kernel 2 Step 8
GPU Cycle=15840734642 | Elapsed=8.00s | Kernel 2 Step 9
GPU Cycle=17820819427 | Elapsed=9.00s | Kernel 2 Step 10
Non-blocking stream 与 default stream 完成
Default stream 与 blocking stream
Kernel 0 started.
Kernel 1 started.
Clock Rate: 1980000000 Hz
Clock Rate: 1980000000 Hz
GPU Cycle=59519 | Elapsed=0.00s | Kernel 0 Step 1
GPU Cycle=62054 | Elapsed=0.00s | Kernel 1 Step 1
GPU Cycle=1980149151 | Elapsed=1.00s | Kernel 1 Step 2
GPU Cycle=1980143082 | Elapsed=1.00s | Kernel 0 Step 2
GPU Cycle=3960237067 | Elapsed=2.00s | Kernel 1 Step 3
GPU Cycle=3960227151 | Elapsed=2.00s | Kernel 0 Step 3
```

```
GPU Cycle=5940325755 | Elapsed=3.00s | Kernel 1 Step 4
GPU Cycle=5940311620 | Elapsed=3.00s | Kernel 0 Step 4
GPU Cycle=7920414092 | Elapsed=4.00s | Kernel 1 Step 5
GPU Cycle=7920396707 | Elapsed=4.00s | Kernel 0 Step 5
GPU Cycle=9900502395 | Elapsed=5.00s | Kernel 1 Step 6
GPU Cycle=9900481204 | Elapsed=5.00s | Kernel 0 Step 6
GPU Cycle=11880590541 | Elapsed=6.00s | Kernel 1 Step 7
GPU Cycle=11880565467 | Elapsed=6.00s | Kernel 0 Step 7
GPU Cycle=13860679020 | Elapsed=7.00s | Kernel 1 Step 8
GPU Cycle=13860649875 | Elapsed=7.00s | Kernel 0 Step 8
GPU Cycle=15840767152 | Elapsed=8.00s | Kernel 1 Step 9
GPU Cycle=15840734691 | Elapsed=8.00s | Kernel 0 Step 9
GPU Cycle=17820856018 | Elapsed=9.00s | Kernel 1 Step 10
GPU Cycle=17820819010 | Elapsed=9.00s | Kernel 0 Step 10
Kernel 2 started.
Clock Rate: 1980000000 Hz
GPU Cycle=58641 | Elapsed=0.00s | Kernel 2 Step 1
GPU Cycle=1980142341 | Elapsed=1.00s | Kernel 2 Step 2
GPU Cycle=3960226862 | Elapsed=2.00s | Kernel 2 Step 3
GPU Cycle=5940311273 | Elapsed=3.00s | Kernel 2 Step 4
GPU Cycle=7920396034 | Elapsed=4.00s | Kernel 2 Step 5
GPU Cycle=9900481086 | Elapsed=5.00s | Kernel 2 Step 6
GPU Cycle=11880565962 | Elapsed=6.00s | Kernel 2 Step 7
GPU Cycle=13860651949 | Elapsed=7.00s | Kernel 2 Step 8
GPU Cycle=15840736681 | Elapsed=8.00s | Kernel 2 Step 9
GPU Cycle=17820821816 | Elapsed=9.00s | Kernel 2 Step 10
Default stream 与 blocking stream 完成
Default stream 与 non-blocking stream
Kernel 0 started.
Kernel 1 started.
Clock Rate: 1980000000 Hz
Clock Rate: 1980000000 Hz
GPU Cycle=60298 | Elapsed=0.00s | Kernel 0 Step 1
GPU Cycle=62663 | Elapsed=0.00s | Kernel 1 Step 1
GPU Cycle=1980152108 | Elapsed=1.00s | Kernel 1 Step 2
GPU Cycle=1980144300 | Elapsed=1.00s | Kernel 0 Step 2
GPU Cycle=3960240963 | Elapsed=2.00s | Kernel 1 Step 3
GPU Cycle=3960228583 | Elapsed=2.00s | Kernel 0 Step 3
GPU Cycle=5940329929 | Elapsed=3.00s | Kernel 1 Step 4
GPU Cycle=5940313735 | Elapsed=3.00s | Kernel 0 Step 4
GPU Cycle=7920419514 | Elapsed=4.00s | Kernel 1 Step 5
GPU Cycle=7920398584 | Elapsed=4.00s | Kernel 0 Step 5
GPU Cycle=9900508385 | Elapsed=5.00s | Kernel 1 Step 6
GPU Cycle=9900483749 | Elapsed=5.00s | Kernel 0 Step 6
GPU Cycle=11880597625 | Elapsed=6.00s | Kernel 1 Step 7
GPU Cycle=11880567770 | Elapsed=6.00s | Kernel 0 Step 7
GPU Cycle=13860686965 | Elapsed=7.00s | Kernel 1 Step 8
GPU Cycle=13860653108 | Elapsed=7.00s | Kernel 0 Step 8
```

```
GPU Cycle=15840775676 | Elapsed=8.00s | Kernel 1 Step 9
GPU Cycle=15840737857 | Elapsed=8.00s | Kernel 0 Step 9
GPU Cycle=17820864369 | Elapsed=9.00s | Kernel 1 Step 10
GPU Cycle=17820822581 | Elapsed=9.00s | Kernel 0 Step 10
Kernel 2 started.
Clock Rate: 1980000000 Hz
GPU Cycle=58124 | Elapsed=0.00s | Kernel 2 Step 1
GPU Cycle=1980142045 | Elapsed=1.00s | Kernel 2 Step 2
GPU Cycle=3960226697 | Elapsed=2.00s | Kernel 2 Step 3
GPU Cycle=5940311420 | Elapsed=3.00s | Kernel 2 Step 4
GPU Cycle=7920396664 | Elapsed=4.00s | Kernel 2 Step 5
GPU Cycle=9900481533 | Elapsed=5.00s | Kernel 2 Step 6
GPU Cycle=11880566722 | Elapsed=6.00s | Kernel 2 Step 7
GPU Cycle=13860651747 | Elapsed=7.00s | Kernel 2 Step 8
GPU Cycle=15840736512 | Elapsed=8.00s | Kernel 2 Step 9
GPU Cycle=17820821152 | Elapsed=9.00s | Kernel 2 Step 10
Default stream 与 non-blocking stream 完成
Non-blocking stream 与 blocking stream
Kernel 0 started.
Kernel 1 started.
Clock Rate: 1980000000 Hz
Clock Rate: 1980000000 Hz
GPU Cycle=58097 | Elapsed=0.00s | Kernel 0 Step 1
GPU Cycle=63414 | Elapsed=0.00s | Kernel 1 Step 1
GPU Cycle=1980150341 | Elapsed=1.00s | Kernel 1 Step 2
GPU Cycle=1980141718 | Elapsed=1.00s | Kernel 0 Step 2
GPU Cycle=3960238588 | Elapsed=2.00s | Kernel 1 Step 3
GPU Cycle=3960225024 | Elapsed=2.00s | Kernel 0 Step 3
GPU Cycle=5940326921 | Elapsed=3.00s | Kernel 1 Step 4
GPU Cycle=5940309386 | Elapsed=3.00s | Kernel 0 Step 4
GPU Cycle=7920415633 | Elapsed=4.00s | Kernel 1 Step 5
GPU Cycle=7920394607 | Elapsed=4.00s | Kernel 0 Step 5
GPU Cycle=9900504399 | Elapsed=5.00s | Kernel 1 Step 6
GPU Cycle=9900479352 | Elapsed=5.00s | Kernel 0 Step 6
GPU Cycle=11880593013 | Elapsed=6.00s | Kernel 1 Step 7
GPU Cycle=11880563764 | Elapsed=6.00s | Kernel 0 Step 7
GPU Cycle=13860680266 | Elapsed=7.00s | Kernel 1 Step 8
GPU Cycle=13860648411 | Elapsed=7.00s | Kernel 0 Step 8
GPU Cycle=15840768722 | Elapsed=8.00s | Kernel 1 Step 9
GPU Cycle=15840733474 | Elapsed=8.00s | Kernel 0 Step 9
GPU Cycle=17820857892 | Elapsed=9.00s | Kernel 1 Step 10
GPU Cycle=17820818396 | Elapsed=9.00s | Kernel 0 Step 10
Kernel 2 started.
Clock Rate: 1980000000 Hz
GPU Cycle=58275 | Elapsed=0.00s | Kernel 2 Step 1
GPU Cycle=1980141823 | Elapsed=1.00s | Kernel 2 Step 2
GPU Cycle=3960226478 | Elapsed=2.00s | Kernel 2 Step 3
GPU Cycle=5940311270 | Elapsed=3.00s | Kernel 2 Step 4
```

```
GPU Cycle=7920396137 | Elapsed=4.00s | Kernel 2 Step 5
GPU Cycle=9900480678 | Elapsed=5.00s | Kernel 2 Step 6
GPU Cycle=11880565273 | Elapsed=6.00s | Kernel 2 Step 7
GPU Cycle=13860650374 | Elapsed=7.00s | Kernel 2 Step 8
GPU Cycle=15840735242 | Elapsed=8.00s | Kernel 2 Step 9
GPU Cycle=17820820396 | Elapsed=9.00s | Kernel 2 Step 10
Non-blocking stream 与 blocking stream 完成
Thread 0 finished.
```

3. synchronize stream

`cmake -B build && cmake --build build`

`./build/csrc/cuda_stream/synchronize_stream`

3.1. legacy default stream

result C++

```
$./build/csrc/cuda_stream/synchronize_stream
Legacy default stream
Starting two threads...
Using default stream handle: 0
Behavior depends on compilation flag.

Synchronize legacy default stream
Kernel 0 started.
Clock Rate: 1980000000 Hz
GPU Cycle=60359 | Elapsed=0.00s | Kernel 0 Step 1
GPU Cycle=1980145917 | Elapsed=1.00s | Kernel 0 Step 2
GPU Cycle=3960243314 | Elapsed=2.00s | Kernel 0 Step 3
GPU Cycle=5940341097 | Elapsed=3.00s | Kernel 0 Step 4
GPU Cycle=7920439657 | Elapsed=4.00s | Kernel 0 Step 5
GPU Cycle=9900539025 | Elapsed=5.00s | Kernel 0 Step 6
GPU Cycle=11880954737 | Elapsed=6.00s | Kernel 0 Step 7
GPU Cycle=13861054075 | Elapsed=7.00s | Kernel 0 Step 8
GPU Cycle=15841672163 | Elapsed=8.00s | Kernel 0 Step 9
GPU Cycle=17821771427 | Elapsed=9.00s | Kernel 0 Step 10
Kernel 1 started.
Clock Rate: 1980000000 Hz
GPU Cycle=60219 | Elapsed=0.00s | Kernel 1 Step 1
GPU Cycle=1980145086 | Elapsed=1.00s | Kernel 1 Step 2
GPU Cycle=3960242470 | Elapsed=2.00s | Kernel 1 Step 3
GPU Cycle=5940339252 | Elapsed=3.00s | Kernel 1 Step 4
GPU Cycle=7920428027 | Elapsed=4.00s | Kernel 1 Step 5
GPU Cycle=9900515878 | Elapsed=5.00s | Kernel 1 Step 6
GPU Cycle=11880603900 | Elapsed=6.00s | Kernel 1 Step 7
GPU Cycle=13860692167 | Elapsed=7.00s | Kernel 1 Step 8
GPU Cycle=15840780099 | Elapsed=8.00s | Kernel 1 Step 9
GPU Cycle=17820867946 | Elapsed=9.00s | Kernel 1 Step 10
Kernel 2 started.
Clock Rate: 1980000000 Hz
GPU Cycle=60351 | Elapsed=0.00s | Kernel 2 Step 1
GPU Cycle=1980145429 | Elapsed=1.00s | Kernel 2 Step 2
GPU Cycle=3961073728 | Elapsed=2.00s | Kernel 2 Step 3
GPU Cycle=5941171208 | Elapsed=3.00s | Kernel 2 Step 4
GPU Cycle=7921269442 | Elapsed=4.00s | Kernel 2 Step 5
GPU Cycle=9901367588 | Elapsed=5.00s | Kernel 2 Step 6
GPU Cycle=11881465625 | Elapsed=6.00s | Kernel 2 Step 7
GPU Cycle=13861563887 | Elapsed=7.00s | Kernel 2 Step 8
GPU Cycle=15841662990 | Elapsed=8.00s | Kernel 2 Step 9
GPU Cycle=17821761107 | Elapsed=9.00s | Kernel 2 Step 10
Synchronize legacy default stream completed
```

3.2. blocking stream

result C++

```
Synchronize blocking stream
Kernel 0 started.
Clock Rate: 1980000000 Hz
GPU Cycle=60717 | Elapsed=0.00s | Kernel 0 Step 1
GPU Cycle=1980146224 | Elapsed=1.00s | Kernel 0 Step 2
GPU Cycle=3960243383 | Elapsed=2.00s | Kernel 0 Step 3
GPU Cycle=5941655261 | Elapsed=3.00s | Kernel 0 Step 4
GPU Cycle=7921754487 | Elapsed=4.00s | Kernel 0 Step 5
GPU Cycle=9901852336 | Elapsed=5.00s | Kernel 0 Step 6
GPU Cycle=11881951559 | Elapsed=6.00s | Kernel 0 Step 7
GPU Cycle=13862050889 | Elapsed=7.00s | Kernel 0 Step 8
GPU Cycle=15842150399 | Elapsed=8.00s | Kernel 0 Step 9
GPU Cycle=17822250773 | Elapsed=9.00s | Kernel 0 Step 10
Kernel 1 started.
Clock Rate: 1980000000 Hz
GPU Cycle=60489 | Elapsed=0.00s | Kernel 1 Step 1
GPU Cycle=1980146796 | Elapsed=1.00s | Kernel 1 Step 2
GPU Cycle=3960244225 | Elapsed=2.00s | Kernel 1 Step 3
GPU Cycle=5940697330 | Elapsed=3.00s | Kernel 1 Step 4
GPU Cycle=7920796287 | Elapsed=4.00s | Kernel 1 Step 5
GPU Cycle=9900895290 | Elapsed=5.00s | Kernel 1 Step 6
GPU Cycle=11880994947 | Elapsed=6.00s | Kernel 1 Step 7
GPU Cycle=13861094589 | Elapsed=7.00s | Kernel 1 Step 8
GPU Cycle=15841194359 | Elapsed=8.00s | Kernel 1 Step 9
GPU Cycle=17821294291 | Elapsed=9.00s | Kernel 1 Step 10
Synchronize blocking stream completed
Kernel 2 started.
Clock Rate: 1980000000 Hz
GPU Cycle=60414 | Elapsed=0.00s | Kernel 2 Step 1
GPU Cycle=1980145282 | Elapsed=1.00s | Kernel 2 Step 2
GPU Cycle=3960241879 | Elapsed=2.00s | Kernel 2 Step 3
GPU Cycle=5940338690 | Elapsed=3.00s | Kernel 2 Step 4
GPU Cycle=7920437872 | Elapsed=4.00s | Kernel 2 Step 5
GPU Cycle=9900536506 | Elapsed=5.00s | Kernel 2 Step 6
GPU Cycle=11880635243 | Elapsed=6.00s | Kernel 2 Step 7
GPU Cycle=13861324597 | Elapsed=7.00s | Kernel 2 Step 8
GPU Cycle=15841423507 | Elapsed=8.00s | Kernel 2 Step 9
GPU Cycle=17822367853 | Elapsed=9.00s | Kernel 2 Step 10
Thread 0 finished.
```

Reference

CUDA stream behavior

CUDA 知识点：流与并发执行 | CUDA

llm-inference-practice/csrc/cuda_stream at main · yykzjh/llm-inference-practice